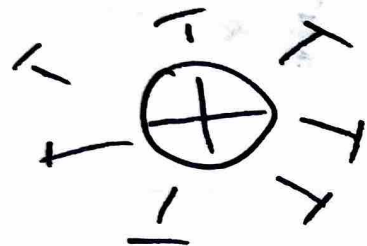


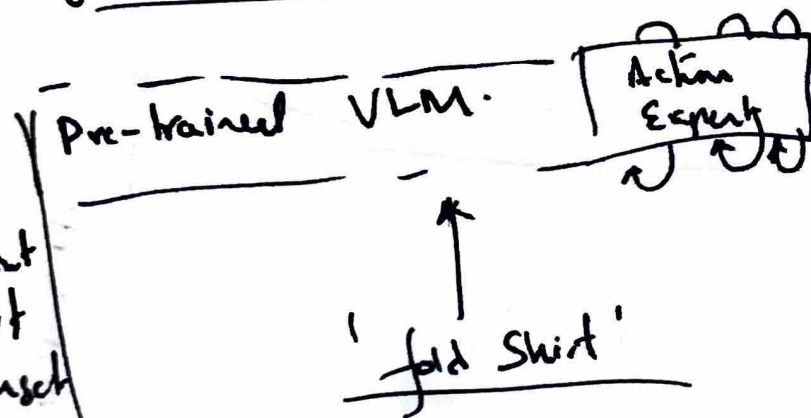
VLA π_0 Model

→ Internet Scale training



OPEN-X Embedment

π_0 V-L-A - Model



Model

No vision - Action learning

Questions

1. Where is this dataset procured from?
2. Where is the high quality Post-training data?

→ High dexterity tasks

→ Unseen tasks

Recipe

- must be done @ large scale
- Right model architectures
- Right recipes.

- Flow matching : (Action Expert, Model)

- variant $\uparrow$ Diffusion
- Cross Embodiment Training $\otimes$
- Control robots upto 50Hz

(Features) Action Chunking

TRAINING PIPELINE / RECIPE

- First train on large & diverse corpus
- fire-tuned on specific data
- * Dexterity
- * Efficiency
- * Robustness.

Data	
High quality	Low quality
$\downarrow$	$\downarrow$
+ Derived behavior patterns	+ Act robustly
- mistakes not seen	- Efficiently
High Quality	
Pretraining over 10000 hours Robot data.	

- fine-tuning
- dexterous tasks
- Laundry
 - cleaning table.
 - putting dishes away.
 - Stack eggs - Cans
 - bagging groceries.

Auto Regressive discretize

Flow DECODER
~~Diffusion~~

$\downarrow$
Cross Entropy loss

DIFFUSION

$\downarrow$
Flow matching

OXE dataset

- long tasks
- 10x more length
- combine physical dexterity & complexity

Magic Soup

- Pretraining Phase (x)
- Post-training Phase (xx)

"PaliGenema"

Pretrained VLM	
(gpt LIP) Genema	
400M	2.6B

Embeddings

"Flow matcher"

Action

360M

Expert

Noise injected

Embeddings

view can be any

⇔ SWAP PALIGENEMA
for other as well.

M.O. Experts

- (1) Proprioceptive State (1)
- (2) Robot action (2)

Mixture of experts

- Robot sp. Action
- Action Expert.

$$p(A_t | o_t)$$

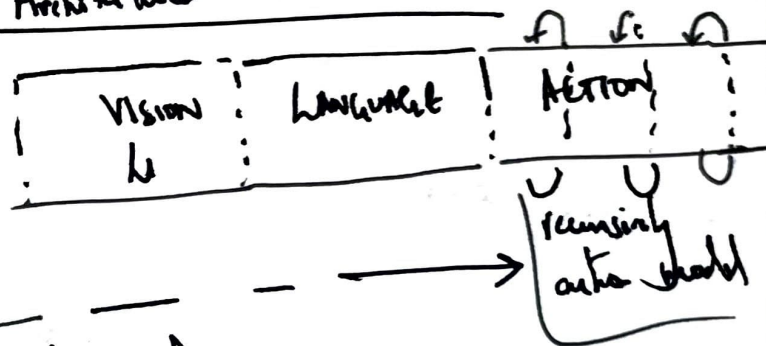
$$A_t = \begin{bmatrix} a_t, a_{t+1}, a_{t+2} \dots \\ a_{t+H-1} \end{bmatrix}$$

where $H = 50$,

$o_t = \text{observed}$

1. RGB image.
2. lang. command.
3. Robot pos. $\rightarrow$

Architecture Model



$(a'_t) \in A_t$
 $A_t: \text{Action chunk}$

$I_t^1 \dots I_t^n$: ith image
 L_t : seq. of lang. tokens
 q_t : joint-angle collection.

$a'_t \in A_t$ (Action chunk)

action token $\rightarrow$ Action exp.

Encoded: Space

Encoded: Space

Encoded: Space

Flow Loss
 Matching

$$\mathcal{L}^z(\theta) = \mathbb{E}_{p(A_t | o_t), q(A_t^z | A_t)}$$

Expected following p & q
 loss is given as:

$$\|V_\theta(A_t^z, o_t) - \mu(A_t^z | A_t)\|^2$$



Known Gaussian
prob path

(flow walking path)

Noise
Data

→ 2:44 pm

Transformation
→ How does it work
Gaussian - Laplace
vs alternative

32
28
20
46
59

Action output

Transformation

REFERENCES

Vector field

$Z = 0, 1$

Adding noise so as to cut input

Noise	High	Low
$Z =$	0	1

$\text{Cov}(0, I)$

$A_t = 2A_t + (1-2) \epsilon$

$A_t^2 = A_t + \epsilon$

Step 3: $u(A_t^2 | A_t) = 3 - A_t$
 $u(A_t^2 | A_t) = (V | u)$
 Low utility & V

3:44 pm
11:00 pm
Code paper
Transformation

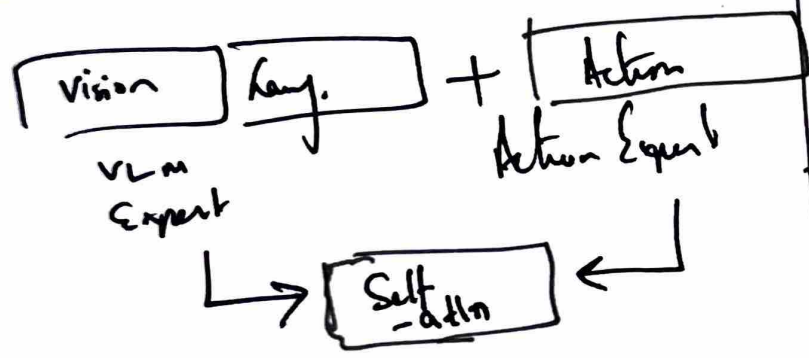
Behaviour 1 & 2

Noise in random sampling
Add noise

new wave of driving sample
found? memory? Doherty discovery

π_θ Pignorch Model

Pseudo Code!?

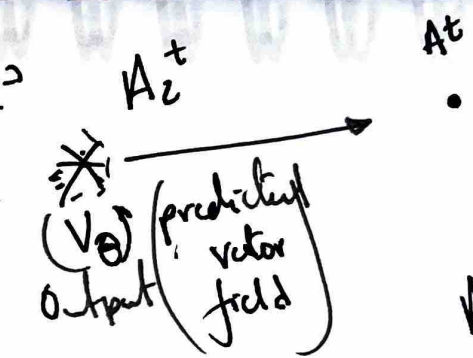


$$A_t^z = z A_t + (1-z) \epsilon$$

$$\mu = \epsilon - A_t \quad (\text{Ideal dir.})$$

Action Expert

A_t^z	z	O_t
Noisy action	time step	obs.
Input		



$$A_t^z + \epsilon - A_t$$

$$A_t^z - A_t = 1 -$$

$$\epsilon =$$

$$A_t^z \leftarrow \delta = A_t^z + \delta V_\theta(A_t^z, O_t)$$

$$(u + A_t)??$$